



EDUCATION

Year	Degree/Exam	Institute	CGPA/Marks
2026	B.TECH	IIT Kharagpur	8.63 / 10
2022	HSC	B.R. Ambedkar Jr. College	89.84%
2020	ICSE	Carmel Academy ICSE, Datala	94%

INTERNSHIPS

Data Analyst Intern | Sabre Corporation

[May '24 - July' 24]

- Reduced the MSE score from **0.0096** to **0.0046** using the Stacked Ensemble AutoML model from H2O.AI, for non-stop share prediction
- Utilized the **NetworkX** library for feature engineering and nodes clustering coefficient calculation of **9928** markets based on **Graph Theory**
- Deployed a **RAG-based** chatbot using **Gemini Pro API** to provide SQL queries, python codes and plots from uploaded CSV and PDF files

Foreign Training Program

[Mar '24 - Aug' 24]

Supervisor: Prof. Gang Li., School of Info. Technology, Deakin University, Australia

- Analyzed **iForest**, **iNNE**, and **IDK** anomaly detection methods to compile a literature review report on **DDoS attack detection** techniques
- Secured **100% accuracy** using a **Deep Residual Network** model with Dice Loss on CSE-CIC-IDS 2018 dataset after SMOTE augmentation
- Achieved maximum **99.47% accuracy** and a **0.99 F1 score** on the CIC-DDos 2019 dataset using **LUCID** and **FLAD** models respectively

Machine Learning Intern | Suvidha Foundation

[Dec '23 - Feb' 24]

- Cleaned training and validation dataset with **83589** and **14750** news articles respectively and utilized **GloVe** embeddings for vectorization
- Implemented a **Pointer Generator network** using **GRU layers** and **NLL loss** function to summarize news articles in human-style language
- Scraped a total of **3357 articles** published on the Press Information Bureau (PIB) website for the quarter from January to March 2024

PROJECTS

ReefChat: Multi-Agent Retrieval System | InterIIT Pathway Problem Statement

- Developed Multi-agent Orchestrated Retrieval and DAG sYntesis(MORAY) to enable parallelized execution of agents and tools using a DAG
- Implemented 2-Stage Multi-Hop Retrieval and Metadata Filtering, achieving an 11% improvement in retrieval accuracy by refining context
- Integrated STELLA-1.5B embedding model leveraging Haiku by Claude 3 for contextual chunking enhancing information retrieval accuracy
- Incorporated NVIDIA NeMo guardrails to enforce ethical AI standards, enhance fault-tolerant workflows, & ensure robust system reliability

Advanced Image Processing and Generation Application | [Notebook Link]

- Developed a **Streamlit** application featuring a custom mask generator, enabling users to upload, process, and enhance uploaded images
- Implemented black-and-white image colorization using **OpenCV** and a **pre-trained Caffe model**, enhancing image quality and detail
- Integrated **Stable Diffusion XL models** for text-to-image and image-to-image generation, producing high-fidelity visuals from user inputs
- Enabled inpainting using the **Kandinsky model** for precise masking and image restoration, improving visual coherence and accuracy

KIK Bot: AI-Powered Personalized Content Generation | [Github Link]

- Developed an AI-powered chatbot using Streamlit, LangChain, and **OpenAI GPT-3.5-turbo** for personalized content generation
- Implemented conversation management with **LangChain's ConversationChain** and **ConversationEntityMemory** for context-awareness
- Designed a user-friendly interface with **Streamlit**, integrating user input collection, conversation history storage, and session management

PUBLICATIONS

Exploring Crowdfunding Campaign Success Using Multimodal Sentiment Analysis

[Dec '23 - June '24]

Authors: Prof. Sobhan S, IIM Ranchi, Bhavesh P | Journal of Organizational Computing and Electronic Commerce | [Manuscript Submitted]

- Employed OpenCV and **Haar cascade classifier** to detect human faces along with **Resnet50 CNN model** for object recognition
- Implemented congruency analysis of cover images and textual description using **Contrastive Language-Image Pre-training (CLIP)** model
- Proposed **Single-Query Rank Boosting model (SQRankBoost)** built on **LambdaMART** for campaign rank prediction in order of trend
- Performed **Mann-Whitney U test** and **Wilcoxon signed-rank tests** to conduct statistical analysis employing a non-parametric approach
- Achieved a top NDCG score of **0.96** with Random Forest along with a DCG score of **1420.32** using the proposed SQRankBoost model

SKILLS AND EXPERTISE

Programming and Web Languages: Proficient - Python | C++ | C | SQL | TeX Intermediate - HTML | CSS | Javascript | Typescript
Frameworks and Libraries: Django | TensorFlow | PyTorch | OpenCV | Langchain | HuggingFace | Selenium | AWS | LangGarph | Autogen
Coursework: Programming & Data Structures (Lab and Theory) | Probability & Statistics | Linear Algebra, Numerical & Complex Analysis

POSITIONS OF RESPONSIBILITY

- Executive Head** of Kharagpur Data Analytics Group, Technology Students Gymkhana, IIT Kharagpur for the academic session 2024-2025
- AI and Metaverse Head** at KodeinKGP: Technology Web 3.0 Society, IIT Kharagpur, for the entire academic session 2024-2025

COMPETITION/CONFERENCE

- Secured **3rd position** in Fin-A-Lytics, National Finance and Analytics Case Challenge 2023 competing against **100 participants** all over India
- Earned **1st place** as the team leader in the Innovate4Swadeshi Hackathon 2024 held by KodeinKGP, among 50 finalists all across India
- Awarded **1st place** as team captain at St. Chavara Mega Expo-2019, a national science exhibition competing against around **450 schools**

EXTRA CURRICULAR ACTIVITIES

Passed **Madhyama Pratham** in **Harmonium** with distinction (**Equivalent to Diploma**) from Akhil Bhartiya Gandharva Mahavidyalaya, Pune